Toward Automated Perception of Pediatric Preoperative Anxiety on a Social Robot: An Observable-Behavior Approach

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Background. Up to 75% of children undergoing surgery experience elevated preoperative anxiety, which predicts harder anesthetic induction, more postoperative pain, and maladaptive behavior after discharge [1]. Assessment today relies on trained observers using the modified Yale Preoperative Anxiety Scale (mYPAS) [2], which is labor-intensive and captures single moments rather than the full preoperative course. Socially assistive robots could provide continuous, unobtrusive monitoring, but most automated approaches infer discrete emotions from facial expressions, a mapping that is neither reliable nor specific, particularly in children [3]. General-purpose vision-language models also lack the specialized knowledge needed for subtle facial cues [4], and slow or mistimed responses frustrate users of LLM-powered robots and shape how the whole interaction is perceived [5, 6]. Consistent with this, our pilot on a social robot (n = 4) found that per-turn vision-language valence estimates labeled 95% of frames as neutral and added more than two seconds of response latency [7].

Objective. To develop a software perception module that estimates preoperative distress in children aged 5 to 12 continuously, using signals that map directly onto the mYPAS rather than onto emotion labels, and that can later run with an off-the-shelf bedside social robot.

Methods. The current prototype runs locally on a laptop and uses its built-in camera and microphone, with no audio or video leaving the machine. It is designed to move to a bedside social robot such as the Reachy Mini in a later phase. Openly available pose, facial action unit, and acoustic feature extractors estimate the five mYPAS domains: activity, vocalization, emotional expressivity, state of arousal, and use of parents. Where available, heart rate from a bedside pulse oximeter is added as an arousal channel. Because heart rate alone correlates only weakly with mYPAS in children [8], physiology complements rather than replaces behavioral signals, following multimodal work that matched human expert agreement in children interacting with robots [9]. Signals are combined into a continuous distress score referenced to each child's own early baseline and displayed on a clinician-facing view. Before any patient testing, the score will be evaluated against blinded human mYPAS ratings of scripted adult role-play sessions spanning calm, restless, and distressed states, with agreement reported as an intraclass correlation alongside latency and signal availability.

Anticipated contribution and next steps. This work offers a low-cost, hardware-agnostic perception layer expressed in a scale clinicians already use. We will present the working prototype, pre-clinical validation results, and a proposed pilot protocol for children in the preoperative holding area, measuring agreement with observer-rated mYPAS and, for ages 7 and older, self-reported anxiety on the CPMAS [10]. We seek clinical partners to conduct this pilot. In the next phase, the distress score will drive a social robot's responses: when a child's distress rises, the robot will offer an interactive game or guide the child through calming techniques such as paced breathing, building on evidence that interactive game distraction reduces anxiety at parental separation [11].

Table 1. System components. All software is openly available for research and runs locally.

Component	Function	Tool
Hardware	Camera and microphone now; bedside social robot in a later phase	Laptop; Reachy Mini (planned)
Body tracking	Activity, restlessness and proximity to parent from pose keypoints	MediaPipe Pose
Facial analysis	Facial action units read as valence and arousal, not emotion labels	OpenFace
Audio analysis	Crying, vocal pitch and loudness, silence	openSMILE
Physiology	Heart rate as a supporting arousal signal	Bedside pulse oximeter
Distress score	Fuses signals into a mYPAS-range score (23.33–100), referenced to each child's baseline	Custom Python
Clinician view	Live display of the score over time	Web dashboard
Reference standard	Blinded human ratings used to validate the score	mYPAS (CPMAS, ages 7+)

Keywords: pediatric preoperative anxiety; socially assistive robotics; affective computing; mYPAS; multimodal perception

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